

- 3 (a)** A student wanted to light a lamp, but only had available a 12 V battery of negligible internal resistance. In order to reduce the battery voltage, he connected the circuit as shown in Fig. 3.1. The maximum value of the resistance of the rheostat XY was $1000\ \Omega$.

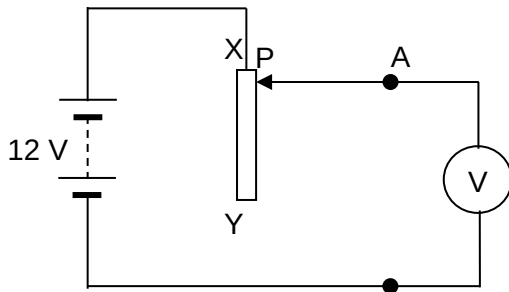


Fig. 3.1

He found that, when the sliding contact P of the rheostat was moved down from X to Y, the voltmeter reading dropped from 12 V to 11 V.

Calculate the resistance of the voltmeter.

		resistance = Ω	[2]
		Let the voltmeter resistance be R_V. When the sliding contact P was at Y, potential divider rule gives: $\left(\frac{R_V}{R_V + 1000} \right) (12 \text{ V}) = 11 \text{ V}$ $\Rightarrow R_V = 11 \text{ k}\Omega$ OR Current through XY = $1/1000 \text{ A}$. $R_V = \frac{11}{1/1000} = 11 \text{ k}\Omega$ Comments: Mostly well done except for sum miscalculation.	
	(b)	He modified the above circuit into the one shown in Fig. 3.2 below, using the rheostat as a potentiometer, and was now able to adjust the rheostat to give a voltmeter reading of 3.0 V. Fig. 3.2	
	(i)	Calculate the current that flows through the voltmeter.	
		current = A	[1]
		$I_V = 3 \text{ V}/11 \text{ k}\Omega$ $= 0.27 \text{ mA}$ Comments: Mostly well done except for sum miscalculation.	
	(ii)	Assuming that the current in (i) is negligible compared with the current through the rheostat, determine how far down from X the sliding contact P would have been moved. Express your answer as a fraction of the length of XY.	

		fraction of the length of XY = A	[2]
		Position of slider has to be such that $\frac{PY}{XY} = \frac{3\text{ V}}{12\text{ V}} = \frac{1}{4}$ Sliding contact would have been moved down 3/4 of its length from X. Comments: Mostly well done except for sum miscalculation.	
	(iii)	The student then removed the voltmeter in Fig.3.2 and then connected a lamp rated at 0.60 W, 3.0 V in its place, but it was very dim. By calculating the power delivered to the lamp, explain this observation.	
			
			
			
			[4]
		resistance of lamp = $V^2/P = (3.0)^2/0.60 = 15\ \Omega$ effective resistance across AB, $R_{AB} = \frac{15 \times 250}{15 + 250} = \underline{14.2\ \Omega}$ p.d. across AB, $V_{AB} = (12) \left(\frac{R_{AB}}{R_{AB} + 750} \right) = \underline{0.222\text{ V}}$ power delivered to the lamp = $\frac{V_{AB}^2}{15\ \Omega} = \underline{0.0033\text{ W}}$ OR current from power supply, $I = \frac{12}{(750 + R_{AB})} = 0.0157\text{ A},$ current through lamp, $I_{\text{lamp}} = I \left(\frac{250}{250 + 15} \right) = 0.0157 \left(\frac{250}{250 + 15} \right) = \underline{0.0148\text{ A}},$ power delivered to the lamp = $I_{\text{lamp}} V_{AB}$ or $I_{\text{lamp}}^2 (15) = \underline{0.0033\text{ W}}$ This is only 0.55% of 0.6 W. Since <u>this is much lower than 0.6 W</u>, its brightness is very low.	

		Comments: Many students have difficulty imagining the effect of the removal of the voltmeter and replacing with a lamp. Because the results that the lamp is very dim is given, students need to be very detailed in explaining.	
			[Total: 9]

4	(a)	(i)	Explain what is meant by the <i>diffraction</i> of a wave	
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